

Statistical Analysis

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```
library(dplyr)
```

Adjuntando el paquete: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
```

```
library(ggplot2)
```

```
library(googlesheets4)
```

```
url <- "https://docs.google.com/spreadsheets/d/1BJR6o_kZH9YCPTlQtPgWaHZdYpI2c1V0J0sdPbb3bPU/"
```

```
df_incidence <- read_sheet(url, sheet = "sampling_sites") |>  
  mutate(across(c(incidence_pct, elevation), as.numeric))
```

! Using an auto-discovered, cached token.

To suppress this message, modify your code or options to clearly consent to the use of a cached token.

See gargle's "Non-interactive auth" vignette for more details:

<<https://gargle.r-lib.org/articles/non-interactive-auth.html>>

i The googlesheets4 package is using a cached token for
'henrysantillanculquimboz@gmail.com'.

v Reading from "dataframe".

v Range 'sampling_sites'.

```
df_nymphs <- read_sheet(url, sheet = "nymph_morphology") |>
  mutate(instar = factor(instar),
         across(c(body_length_mm, body_width_mm), as.numeric))
```

v Reading from "dataframe".

v Range 'nymph_morphology'.

```
df_adults <- read_sheet(url, sheet = "adult_morphology") |>
  mutate(sex = factor(sex))
```

v Reading from "dataframe".

v Range 'adult_morphology'.

```
df_life <- read_sheet(url, sheet = "life_cycle") |>
  mutate(instar = factor(instar))
```

v Reading from "dataframe".

v Range 'life_cycle'.

1 Incidence Analysis

```
cor_pear <- cor.test(~ incidence_pct + elevation, data = df_incidence, method = "pearson")
m1 <- lm(incidence_pct ~ elevation, data = df_incidence)
cor_pear; summary(m1); confint(m1)
```

Pearson's product-moment correlation

```
data: incidence_pct and elevation
t = 3.7944, df = 23, p-value = 0.000936
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.2985353 0.8156369
sample estimates:
      cor
0.6204747
```

```
Call:
lm(formula = incidence_pct ~ elevation, data = df_incidence)
```

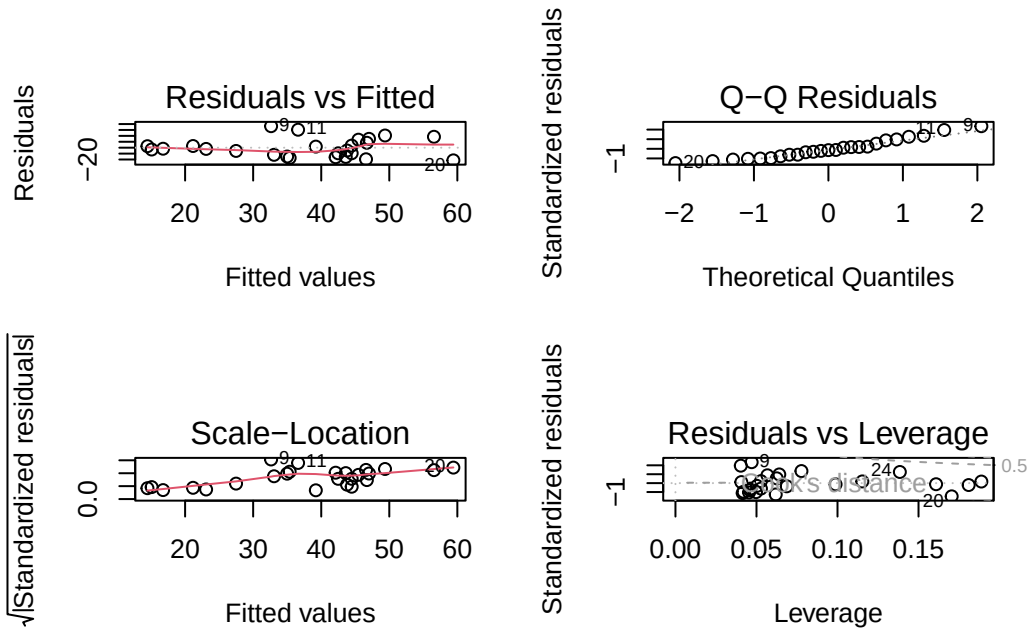
```
Residuals:
      Min       1Q   Median       3Q      Max
-21.418 -12.067  -2.085   8.293  36.367
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.19389    13.52132  -0.902 0.376496
elevation     0.06200     0.01634   3.794 0.000936 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 15.86 on 23 degrees of freedom
Multiple R-squared:  0.385, Adjusted R-squared:  0.3582
F-statistic: 14.4 on 1 and 23 DF, p-value: 0.000936
```

```
              2.5 %      97.5 %
(Intercept) -40.16487096 15.77709889
elevation     0.02819928  0.09580342
```

```
par(mfrow = c(2, 2)); plot(m1); par(mfrow = c(1, 1))
```



```
shapiro.test(residuals(m1)); lmtest::bptest(m1)
```

Shapiro-Wilk normality test

```
data: residuals(m1)
W = 0.94457, p-value = 0.1886
```

studentized Breusch-Pagan test

```
data: m1
BP = 1.4718, df = 1, p-value = 0.2251
```

```
ci <- confint(m1)["elevation", ]
p_txt <- if (cor_pear$p.value < 0.001) "< 0.001" else sprintf("%.3f", cor_pear$p.value)

label_txt <- sprintf(
  "Pearson: r = %.2f, p = %s\nLinear model: \u03B2 = %.3f (95% CI: %.3f\u2013%.3f), R\u00B2
```

```

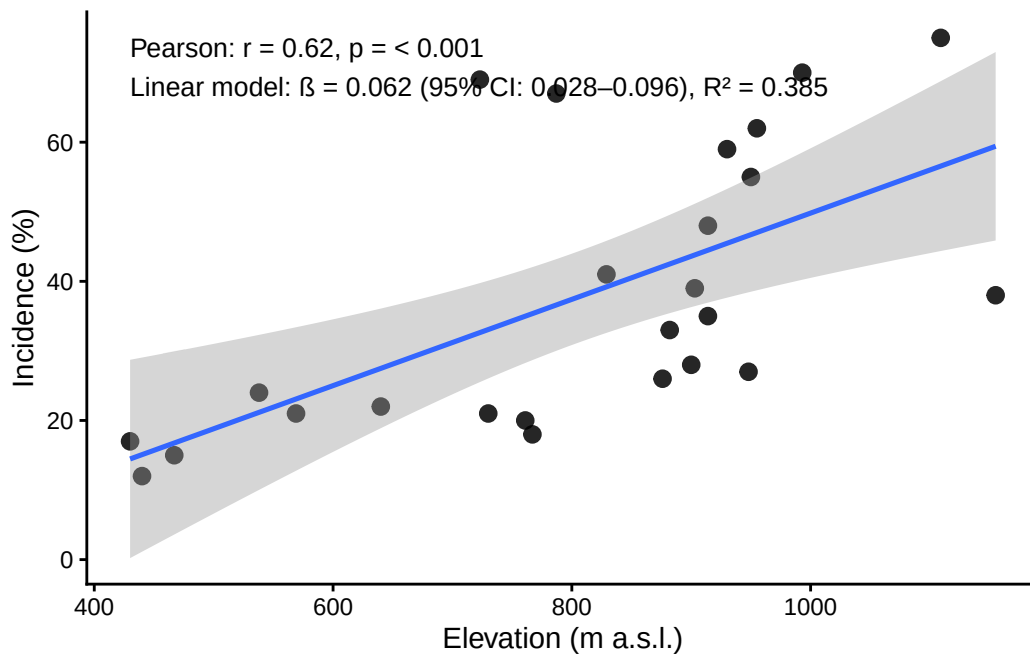
cor_pear$estimate, p_txt, coef(m1)[["elevation"]], ci[1], ci[2], summary(m1)$r.squared
)

p_inc_alt <- ggplot(df_incidence, aes(elevation, incidence_pct)) +
  geom_point(size = 2.6, alpha = 0.85) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 0.9) +
  annotate("text", x = min(df_incidence$elevation), y = max(df_incidence$incidence_pct),
          hjust = 0, vjust = 1, label = label_txt, size = 3.6) +
  labs(x = "Elevation (m a.s.l.)", y = "Incidence (%)") +
  theme_classic()

p_inc_alt

```

`geom\_smooth()` using formula = 'y ~ x'



2 Morphological analyses of nymphal instars

```

table_nymphs <- df_nymphs |>
  pivot_longer(c(body_length_mm, body_width_mm),

```

```

      names_to = "Variable", values_to = "value") |>
group_by(Instar = instar, Variable) |>
summarise(
  n      = sum(!is.na(value)),
  Mean   = mean(value, na.rm = TRUE),
  SD     = sd(value, na.rm = TRUE),
  Min    = min(value, na.rm = TRUE),
  Max    = max(value, na.rm = TRUE),
  CV     = SD / Mean * 100,
  .groups = "drop"
) |>
transmute(
  Instar, Variable, n,
  `Mean ± SD` = sprintf("%.2f ± %.2f", Mean, SD),
  Range       = sprintf("%.2f-%.2f", Min, Max),
  `CV (%)`    = round(CV, 2)
)

table_nymphs

```

A tibble: 10 x 6

	Instar	Variable	n	`Mean ± SD`	Range	`CV (%)`
	<fct>	<chr>	<int>	<chr>	<chr>	<dbl>
1	1	body_length_mm	45	2.00 ± 0.24	1.45-2.45	12.0
2	1	body_width_mm	45	0.88 ± 0.12	0.59-1.09	13.7
3	2	body_length_mm	45	3.38 ± 0.30	2.85-3.98	9
4	2	body_width_mm	45	1.23 ± 0.07	1.10-1.40	6.01
5	3	body_length_mm	45	4.64 ± 0.33	4.09-5.45	7.11
6	3	body_width_mm	45	1.72 ± 0.17	1.45-2.11	9.85
7	4	body_length_mm	45	6.47 ± 0.40	5.80-7.32	6.11
8	4	body_width_mm	45	2.35 ± 0.05	2.24-2.44	2.14
9	5	body_length_mm	45	7.61 ± 0.25	7.20-8.01	3.35
10	5	body_width_mm	45	2.46 ± 0.10	2.30-2.72	3.97

```
writexl::write_xlsx(table_nymphs, "table_nymphs.xlsx")
```

3 Morphological analyses of adults

```

table_adults <- df_adults |>
  pivot_longer(-sex, names_to = "Variable", values_to = "value") |>
  mutate(Variable = factor(Variable, levels = names(df_adults)[-1])) |>
  group_by(Variable, Sex = sex) |>
  summarise(
    n      = sum(!is.na(value)),
    Mean   = mean(value, na.rm = TRUE),
    SD     = sd(value, na.rm = TRUE),
    Min    = min(value, na.rm = TRUE),
    Max    = max(value, na.rm = TRUE),
    CV     = SD / Mean * 100,
    .groups = "drop"
  ) |>
  transmute(
    Variable, Sex, n,
    `Mean ± SD` = sprintf("%.2f ± %.2f", Mean, SD),
    Range       = sprintf("%.2f-%.2f", Min, Max),
    `CV (%)`    = round(CV, 2)
  )

table_adults

```

```

# A tibble: 46 x 6
  Variable      Sex      n `Mean ± SD` Range      `CV (%)`
  <fct>        <fct> <int> <chr>      <chr>      <dbl>
1 body_length female   45 12.89 ± 0.38 11.78-13.66 2.96
2 body_length male     45 10.59 ± 0.39 9.48-11.10 3.71
3 body_width  female   45 2.44 ± 0.11 2.26-2.67 4.36
4 body_width  male     45 2.15 ± 0.11 1.99-2.40 4.99
5 head_length female   45 0.98 ± 0.06 0.81-1.07 5.76
6 head_length male     45 0.87 ± 0.04 0.79-0.93 4.29
7 head_width  female   45 1.66 ± 0.04 1.56-1.73 2.52
8 head_width  male     45 1.47 ± 0.04 1.41-1.53 2.54
9 vertex_width female   45 0.85 ± 0.03 0.79-0.93 4.08
10 vertex_width male     45 0.82 ± 0.06 0.73-0.95 6.77
# i 36 more rows

```

```

writexl::write_xlsx(table_adults, "table_adults.xlsx")

```

4 Analysis of the life cycle

```
table_life <- df_life |>
  mutate(instar = factor(instar,
                        levels = c("egg", "instar I", "instar II", "instar III",
                                   "instar IV", "instar V", "adult"))) |>
  group_by(Stage = instar) |>
  summarise(
    n      = sum(!is.na(duration_days)),
    Mean   = mean(duration_days, na.rm = TRUE),
    SD     = sd(duration_days, na.rm = TRUE),
    Min    = min(duration_days, na.rm = TRUE),
    Max    = max(duration_days, na.rm = TRUE),
    CV     = SD / Mean * 100
  ) |>
  transmute(
    Stage, n,
    `Mean ± SD` = sprintf("%.2f ± %.2f", Mean, SD),
    Range       = sprintf("%d-%d", as.integer(Min), as.integer(Max)),
    `CV (%)`    = round(CV, 2)
  )

table_life
```

```
# A tibble: 7 x 5
  Stage      n `Mean ± SD` Range `CV (%)`
<fct>    <int> <chr>         <chr>    <dbl>
1 egg         47 7.89 ± 1.03 6-10      13
2 instar I    46 4.63 ± 0.53 4-6      11.5
3 instar II   39 4.59 ± 0.50 4-5      10.9
4 instar III  38 4.50 ± 0.51 4-5      11.3
5 instar IV   37 4.49 ± 0.51 4-5      11.3
6 instar V    36 4.50 ± 0.51 4-5      11.3
7 adult      36 9.11 ± 0.57 8-10      6.31
```

```
writexl::write_xlsx(table_life, "table_life.xlsx")
```